nTouch

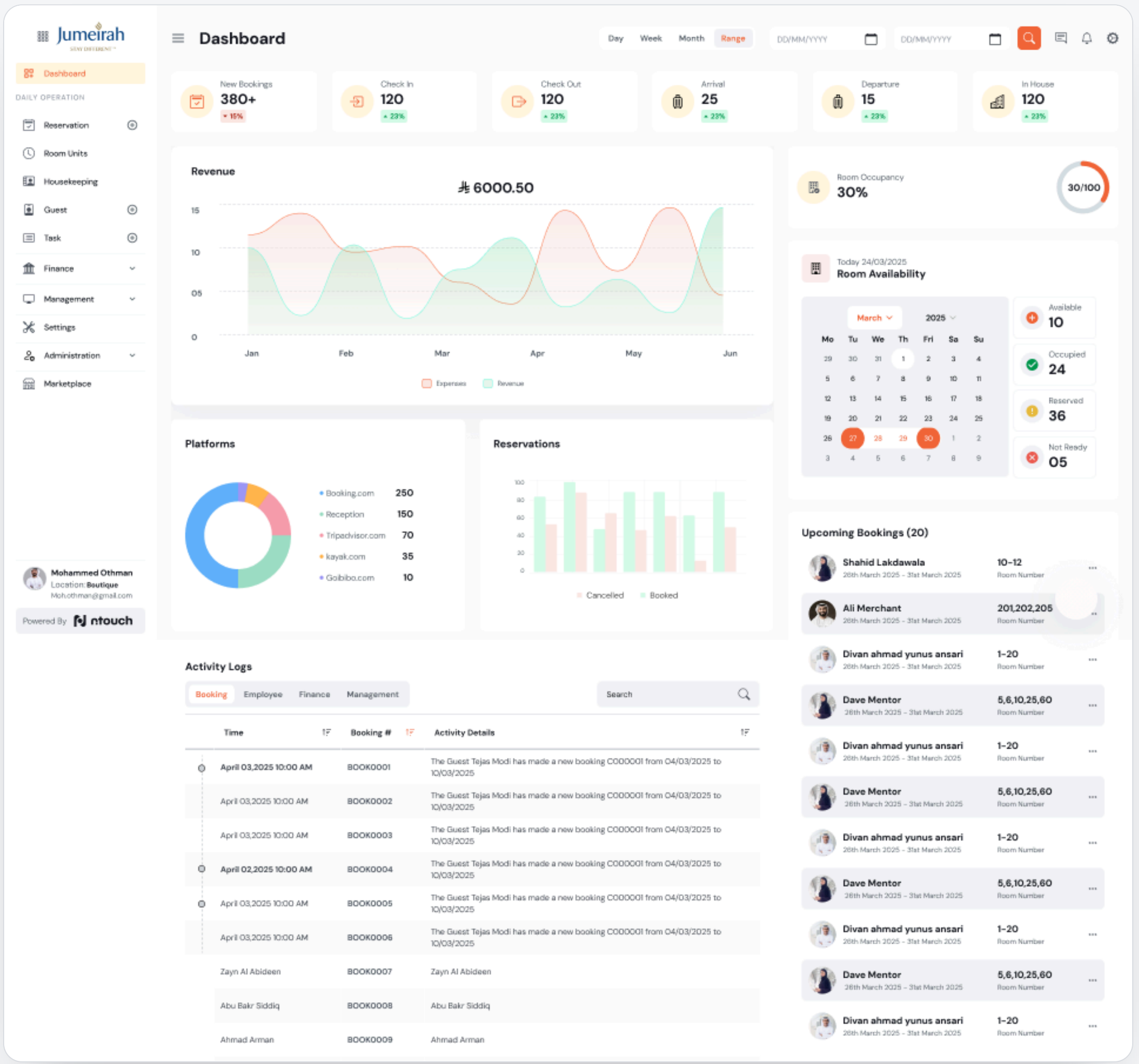
UX Case Study · Hospitality SaaS · 2024–2025

HOTEL PROPERTY MANAGEMENT SYSTEM

Running a hotel from one screen.

nTouch is a cloud property management system for hotels and hotel groups in the GCC. I led research, UX and UI for the web platform and the companion mobile app, from the first stakeholder workshop to the developer handoff.

MY ROLE	PLATFORMS	USERS	MARKET
Lead UX/UI Designer: research, IA, design system, UI, testing, handoff	Web app (1920 desktop) + iOS/Android app	Owners, front desk, housekeeping, finance	Saudi Arabia & UAE, single and multi-property hotels



THE CHALLENGE

Hotels were running on five tools and a whiteboard.

Most mid-size hotels we spoke to used a legacy desktop PMS, a separate channel manager, Excel for cash, WhatsApp for housekeeping and paper for room status. The desk could not see which rooms were actually ready, and owners with several properties waited for end-of-day reports to know how they were doing.

5+

Disconnected tools per hotel

PMS, channel manager, spreadsheets, chat groups and paper logs, each holding part of the truth.

0

Live view of room readiness

Front desk asked housekeeping by phone whether a room was clean before every early check-in.

1 day

Delay on owner reporting

Occupancy, ADR and cash positions arrived as a next-morning PDF, too late to act on.

DESIGN GOAL

Give every role in the hotel one shared, real-time picture of rooms, guests and money, with the next action one click away.

HOW I WORKED

Research to release, in six stages.

The project ran in two-week sprints. Each module (reservations, housekeeping, finance) went through the full loop, so later modules benefited from what earlier tests taught us.

01	02	03	04	05	06
Research	**Define**	**Ideate**	**Design**	**Test**	**Handoff**
Stakeholder workshops, front-desk shadowing, interviews, competitor audit.	Personas, journey maps, problem statements, success metrics.	IA and card sorting, user flows, low-fi wireframes, design studio with devs.	Design system, high-fi web and mobile UI, prototypes.	Moderated usability tests and A/B tests on key flows.	Tokens, specs, states and edge cases, design QA on builds.

01 · RESEARCH

Spending shifts at the front desk.

I combined stakeholder workshops with time on site. Watching a 7 a.m. checkout rush taught us more than any requirement document: the desk, housekeeping and the owner were all asking the same question, "which rooms are ready?", through different channels.

- 14 user interviews
- 3 hotels shadowed
- 4 stakeholder workshops
- 5 competitor PMS audited
- 1 card-sorting study

WORKSHOP NOTES · HOUSEKEEPING

Straight from the Figma working board, after the housekeeping session with the operations team:

1. Staff need to see which rooms to clean, based on upcoming bookings.
2. When housekeeping picks a room, its status should change to **Under cleaning**.
3. Cleaning time should be logged, with a timer to encourage performance.
4. Supervisors want a status screen whose goal is **zero pending rooms**.

Insight 1 **Room status drives everything.** Check-in, upsell and cleaning decisions all depend on knowing whether a room is clean, occupied or out of service right now.

Insight 2 **The desk works in a calendar, not a list.** Agents think in "room × night" and drag stays in their heads; a list view hid gaps and overlaps.

Insight 3 **Owners manage portfolios.** Group owners switch between properties several times a day and want one roll-up view.

Insight 4 **Setup is where hotels quit.** Legacy systems took days of vendor-led setup before the first booking could be made.

Insight 5 **Housekeeping lives on the phone.** Cleaners never sit at a desk, so their tools must be mobile-first and one-thumb.

Competitive audit

PRODUCT	DRAG-AND-DROP CALENDAR	LIVE HOUSEKEEPING	MULTI-PROPERTY SWITCH	SELF-SERVE SETUP	ARABIC / GCC READY
Oracle Opera	Partial	Yes	Yes	No	Partial
Cloudbeds	Yes	Partial	Partial	Yes	No
Mews	Yes	Yes	Yes	Partial	No
Stayntouch	Partial	Yes	Yes	Partial	No
nTouch (target)	Yes	Yes	Yes	Yes	Yes

02 · DEFINE

Four people, one building.

Personas were built from interview clusters. Each one maps to a permission role in the product, which later made access control and dashboard defaults straightforward for the developers.

Mohammed Othman

Owner / General Manager · 46 · Riyadh · 3 properties

"I don't need every button. I need to know by 9 a.m. if today is a good day, and which hotel needs me."

GOALS

- Daily occupancy, ADR and revenue at a glance
- Compare properties
- Spot cash gaps early

FRUSTRATIONS

- Reports arrive the next day
- Logs in to three systems
- Can't audit who changed a rate

Dashboard · Reports · Hotel switcher · Audit logs

Junaid Ahmed

Front Desk Executive · 28 · Jeddah · 8-hour shifts

"When a family walks in at midnight, I have about two minutes to find them a clean room and take payment."

GOALS

- Book a walk-in in under 2 minutes
- See clean, free rooms instantly
- Reuse returning guest details

FRUSTRATIONS

- Calls housekeeping to check rooms
- Double bookings from OTAs
- Re-typing ID details

Reservation calendar · Walk-in booking · Guests

Fatima Khan

Housekeeping Supervisor · 35 · Dubai · team of 12

"My job is to make the dirty-room number zero before the 2 p.m. check-ins start."

GOALS

- Prioritise rooms by next arrival
- Assign tasks to cleaners
- Track cleaning time

FRUSTRATIONS

- Status lives on paper sheets
- No link to bookings
- Out-of-service rooms get sold

Housekeeping · Units · Tasks · Mobile app

Ravi Sharma

Accountant · 39 · Riyadh · shared across hotels

"Every night I reconcile cash box, bank and invoices by hand. One missing receipt costs me an hour."

GOALS

- Close the day without spreadsheets
- VAT-correct invoices
- Export reports for audit

FRUSTRATIONS

- Cash and bank tracked apart
- Discounts not logged
- Manual tax calculation

Cash management · Invoices · Taxes & fees · Reports

02 · DEFINE

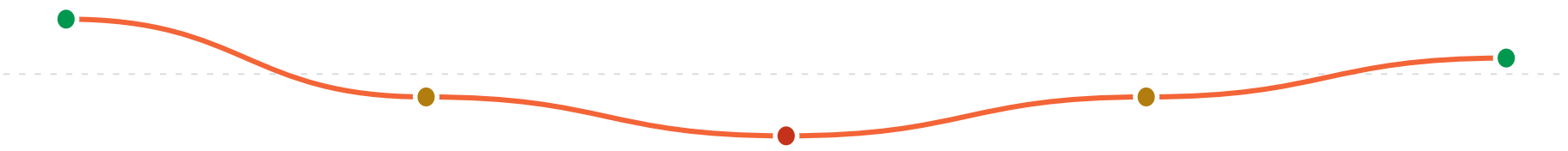
Where each day breaks down.

I mapped the current-state journey for each persona's most frequent critical task. The low points on each curve became the design priorities for that module.

Front desk · Walk-in booking | Housekeeping · Room turnaround | Owner · Morning review | Accountant · Day close

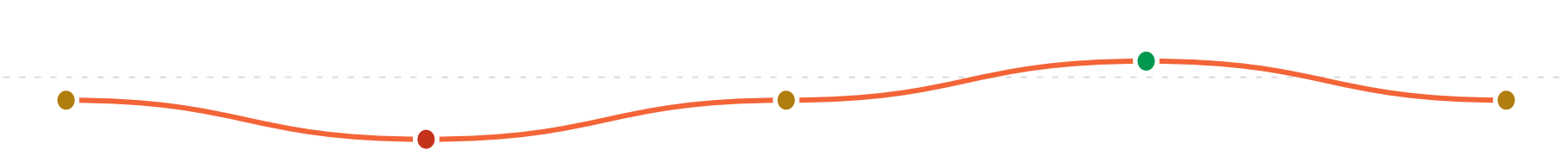
Persona: Junaid, front desk **Scenario:** A family walks in at 11:40 pm without a booking

Goal: Clean room booked and paid in under 2 minutes

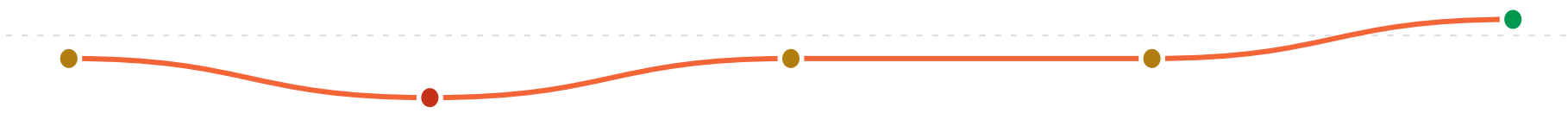
	Greet	Find a room	Check readiness	Enter guest	Take payment
ACTIONS	Greets guests, asks dates and room type	Scans list view across room types	Phones housekeeping to confirm the room is clean	Types name, nationality, ID number by hand	Calculates total with VAT, takes card
THINKING	“How many nights?”	“Is there a twin left?”	“Please be clean...”	“I typed this last month”	“Did I add the tax?”
EMOTION					
PAIN POINTS	—	Gaps hidden in long list	No live room status	No guest lookup	Manual tax and pricing
DESIGN RESPONSE	—	Calendar default with free rooms per day	Clean / not-clean tag on every room	Search existing guest to prefill	Price rules and totals shown before confirm

Persona: Fatima, housekeeping supervisor **Scenario:** Morning checkout rush, 38 rooms to turn before 2 pm

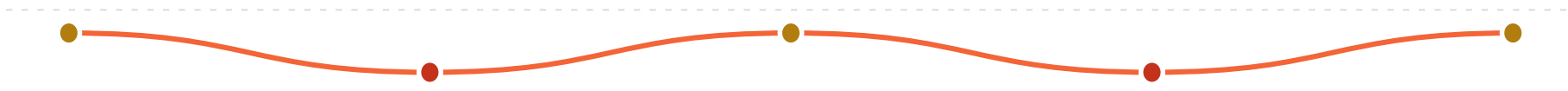
Goal: Zero pending rooms before check-in time

	Get checkouts	Prioritise	Assign	Clean	Report
ACTIONS	Collects printed checkout list from desk	Guesses priority from memory	Shouts and messages room numbers	Cleaners update nothing until done	Calls the desk room by room
THINKING	“Is this list current?”	“Who arrives first?”	“Who's free?”	“How long is 304 taking?”	“Did they get my message?”
EMOTION					
PAIN POINTS	Printed list goes stale	No link to arrivals	Tasks lost in chat	No time tracking	Status lags reality
DESIGN RESPONSE	Live checkout feed	Queue sorted by next arrival	Assign task from the room card	Pick room → Under cleaning + timer	Status syncs to desk instantly

Persona: Mohammed, owner of 3 hotels **Scenario:** First coffee, 8:30 am, before calls start **Goal:** Know which hotel needs attention today

	Open reports	Compare hotels	Check cash	Spot issues	Act
ACTIONS	Opens PDF reports emailed overnight	Opens three logins for three hotels	Asks accountant for cash position	Looks for low occupancy or OTA gaps	Calls the GM of the weak property
THINKING	“This is yesterday's data”	“Why three passwords?”	“Waiting again”	“Who changed this rate?”	“I hope it's not too late”
EMOTION					
PAIN POINTS	Data a day old	No portfolio view	Cash not visible	No audit trail	—
DESIGN RESPONSE	Live dashboard KPIs	Hotel switcher and 'all hotels' view	Cash and bank box on dashboard	Activity and audit logs	Notifications and live chat

Persona: Ravi, accountant **Scenario:** End of day close across two hotels **Goal:** Reconcile cash, bank and invoices without spreadsheets

	Collect receipts	Match payments	Apply taxes	Settle	Export
ACTIONS	Gathers paper receipts from desk	Matches card slips to bookings in Excel	Adds VAT and municipality fee by hand	Counts the cash box	Builds the report for the owner
THINKING	“One is missing again”	“Which booking is this?”	“Did the rule change?”	“Short by 200 SAR”	“Same report every night”
EMOTION					
PAIN POINTS	Paper receipts	No booking link	Manual tax rules	No settlement record	Manual reporting
DESIGN RESPONSE	Receipts logged at payment	Payment number tied to booking	Taxes and fees configured once	Cash box and bank box settlement	One-click export

03 · IDEATE

Navigation that follows the shift.

An open card sort with 11 participants grouped 40+ features. Daily tasks clustered tightly, so they sit at the top of the sidebar; setup and admin moved into collapsible groups so they stay out of the way during a busy shift.

Daily operation
- Dashboard
- Reservation
- Room units
- Housekeeping
- Guest
- Task

Finance
- Cash management
- Invoices
- Vouchers
- Contracts
- Discounts
- Taxes & fees

Management
- Reports
- Activity log
- Documents
- Dynamic pricing
- Channel manager

Settings
- Reservation settings
- Facility & rooms
- Text templates
- Language
- Branding

Administration
- Users & permissions
- Audit logs
- Subscription
- Integrations
- Marketplace

04 · DESIGN SYSTEM

A system built for dense, live data.

The foundations were set up first in Figma: color roles, a type scale on a 1.2 ratio, buttons and inputs with every state, and a single icon family. Each status in the hotel (clean, dirty, occupied, out of service) has one color, used the same way on every screen.

Foundation v1 · #2871E6
Neutral blue used while the structure and components were validated.

→

Product v2 · #F36437
Aligned with the nTouch logo; warm, hospitable, and distinct from blue-heavy competitors.

BRAND & ACTION STATES

Primary	Hover	Pressed	Primary bg	Mobile dark
#F36437	#E0542A	#C2441C	#FDE8DF	#0F2A26

SEMANTIC (STROKE / FILL / TEXT)

Success · Clean	Warning · Dirty	Error · Occupied	Info · Cleaning
#98C729 / #EEFFC5	#E8B93F / #FFEFC7	#DD331D / #F8CEC9	#2C68C6 / #EDF4FF

TYPE COLORS & GRAY SCALE

Heading	Body	Placeholder	Gray 200	Gray 100 · fields
#14181F	#414750	#8B939F	#D0D4D9	#F3F4F5

Typography · DM Sans, 1.2 scale

H1	Room occupancy	Bold 57.3 / 80
H2	Upcoming bookings	Bold 47.8 / 72
H3	Housekeeping	Medium 39.8 / 56
H5	Guest information	Medium 23 / 32
H6	Selected room	Medium 19.2 / 24
Body M	Check-in and checkout are between 12:00 pm and 11:59 am.	Regular 16 / 24
Body S	Super Delux Room 1 · Booking B0012345	Regular 13.3 / 16
Caption	26th March 2025 – 31st March 2025	Regular 11.1 / 16

Buttons · 3 sizes × 5 states

Large 48, medium 40, small 32. Icon left, right, or icon-only variants for each.

Default | Hover | Pressed | Disabled | Secondary

Inputs · default, focus, error

Hotel name

Enter your hotel name

Timezone

United Arab Emirates (GMT+4)

Email

ahmad.ansari@

Enter a full email address, like name@hotel.com

Status tags

Every room and booking state has one tag, used on web and mobile.

Checked in | Reserved | Available | Out of service | Cleaning | Dirty

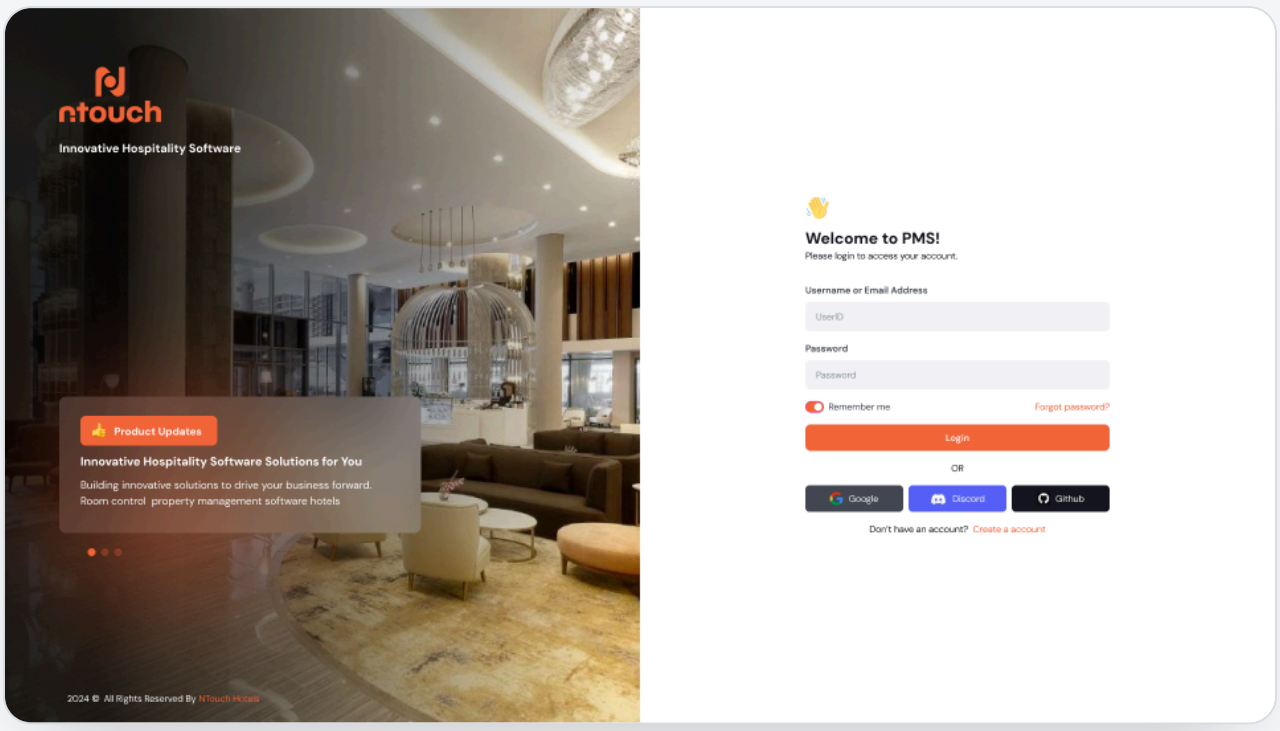
Icons & spacing

Bootstrap Icons at 16 and 24 px for consistency and an open licence developers already knew. Spacing on an 8-point grid; cards use 16 px radius on web, 20 px on mobile.

04 · DESIGN

Key screens and the decisions behind them.

Every screen is designed at 1920 × 1080 on a 290 px sidebar layout. Select any screen to view it larger.



SIGN IN & ONBOARDING

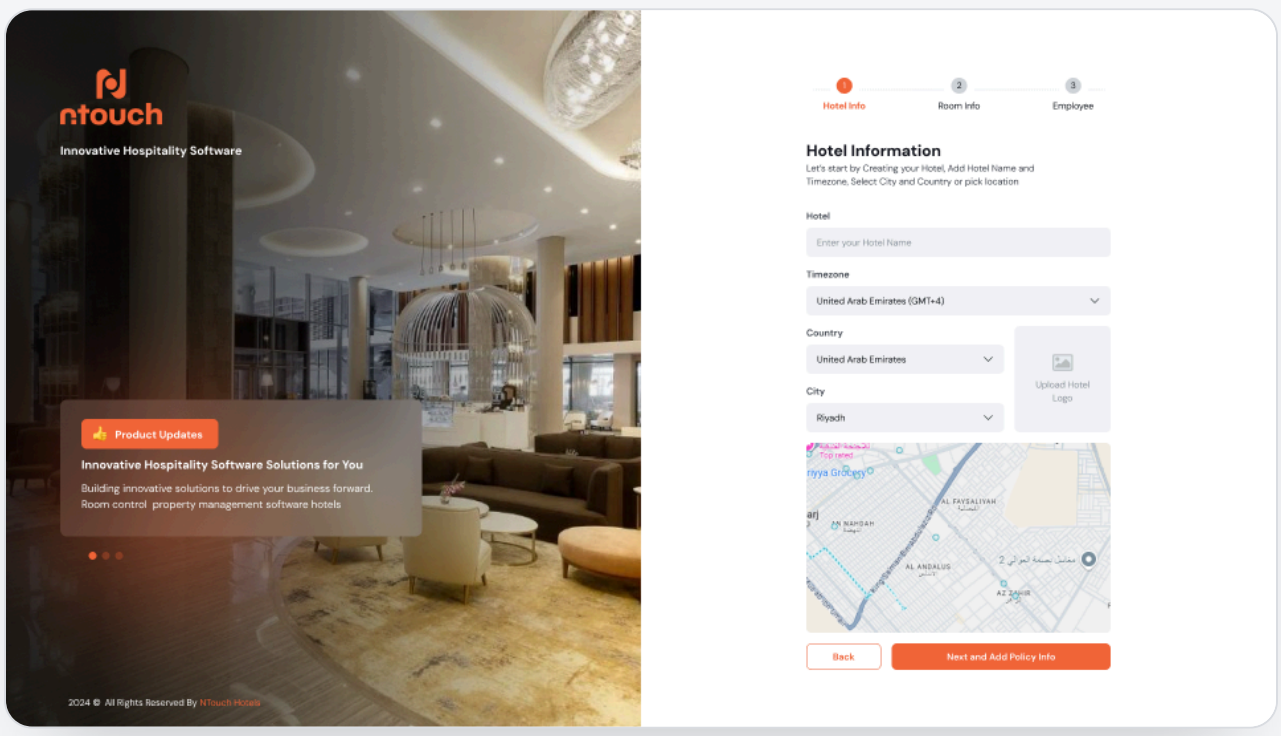
From signup to first booking in one sitting.

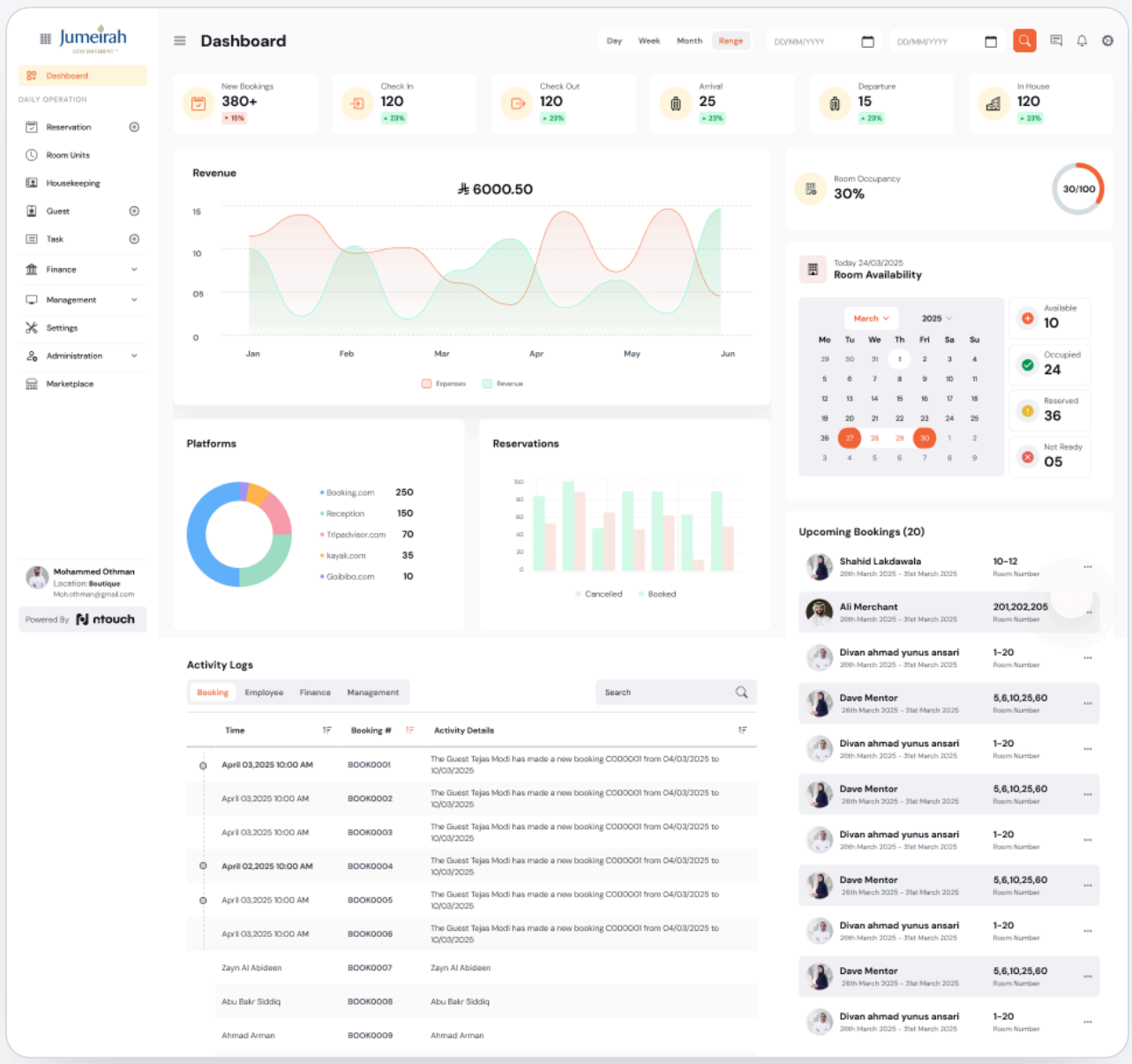
- **Split layout** keeps the form calm while the left panel carries product updates.
- **Tenant-aware login:** a hotel picker appears only when a user belongs to more than one property.
- Clear validation and OTP-based password reset.

GUIDED SETUP

Three steps replace a vendor-led setup.

- **Hotel info → Room info → Employee**, with policy and tax info added after.
- Map pin and timezone defaults from the chosen country.
- Room categories as reusable cards, so a 120-room hotel is set up by category, not room by room.





DASHBOARD

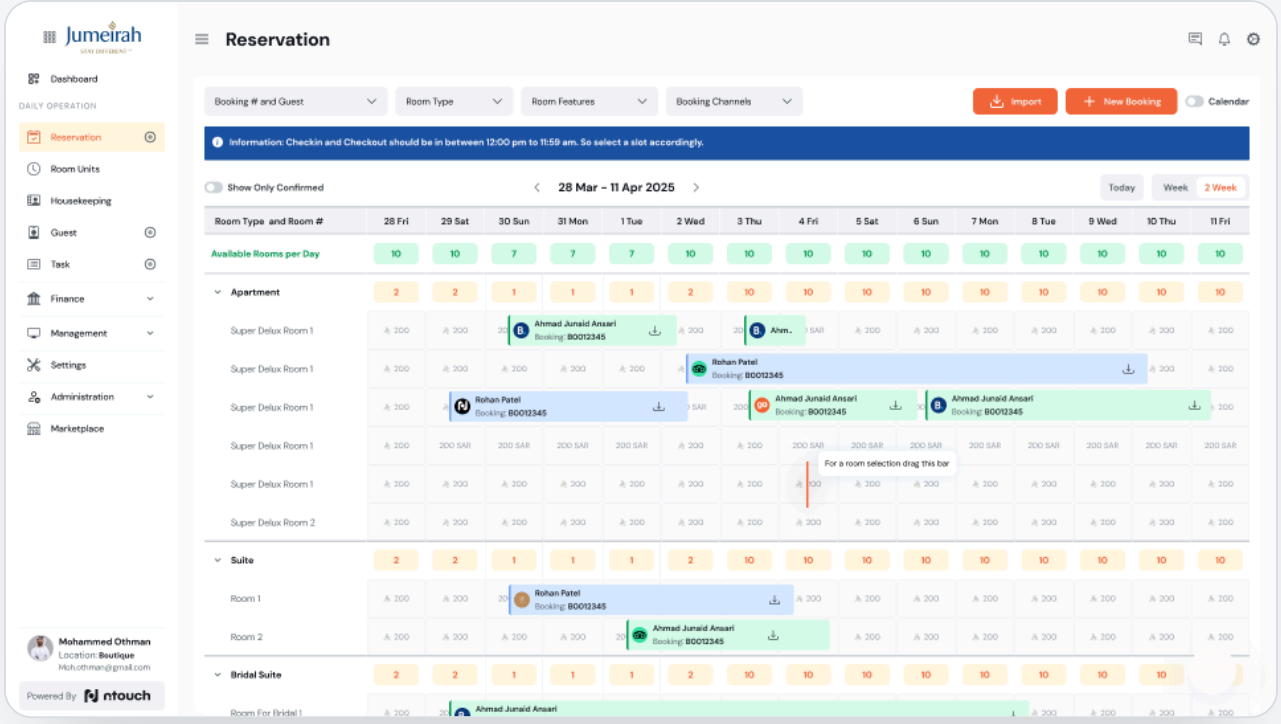
The 9 a.m. question answered above the fold.

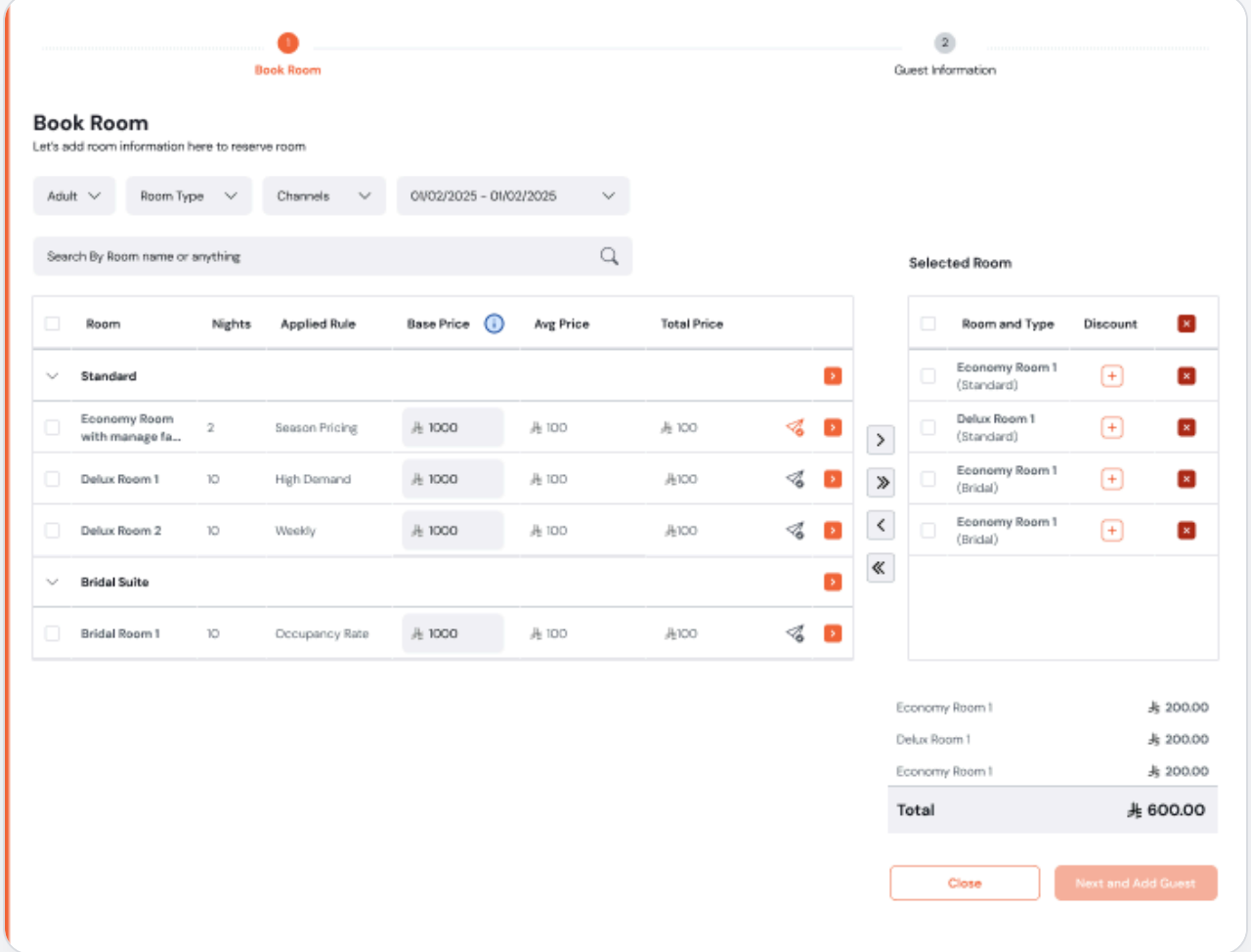
- **Six KPI tiles** with change versus the last period: bookings, check-ins, check-outs, arrivals, departures, in-house.
- Room availability calendar paired with available, occupied, reserved and not-ready counts.
- Activity log tabs (booking, employee, finance) for accountability.

RESERVATION CALENDAR

Room × night, the way agents already think.

- **Drag to select** a range of nights on a room to start a booking.
- Rooms grouped by type, with available rooms per day on top.
- Channel logos (Booking.com, Tripadvisor, direct) on every stay bar.
- Policy banner reminds staff of check-in and checkout windows.





WALK-IN BOOKING

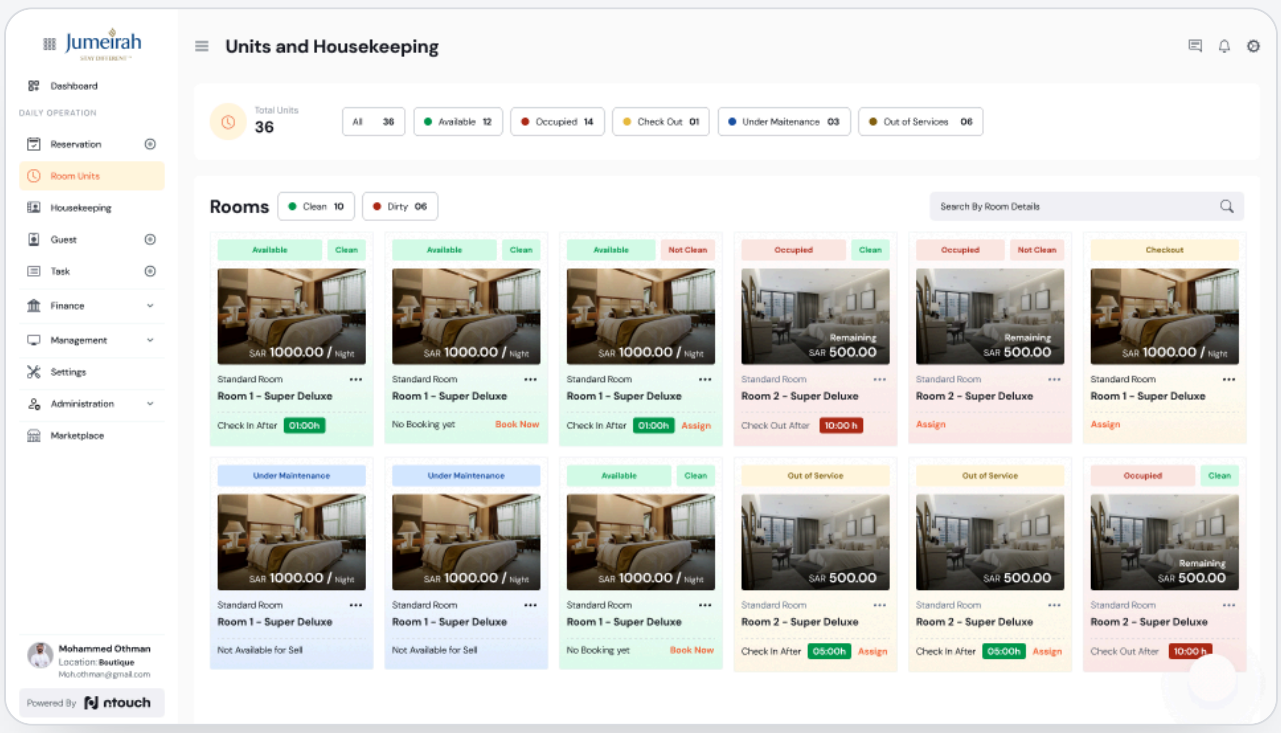
Two steps, with pricing shown before commitment.

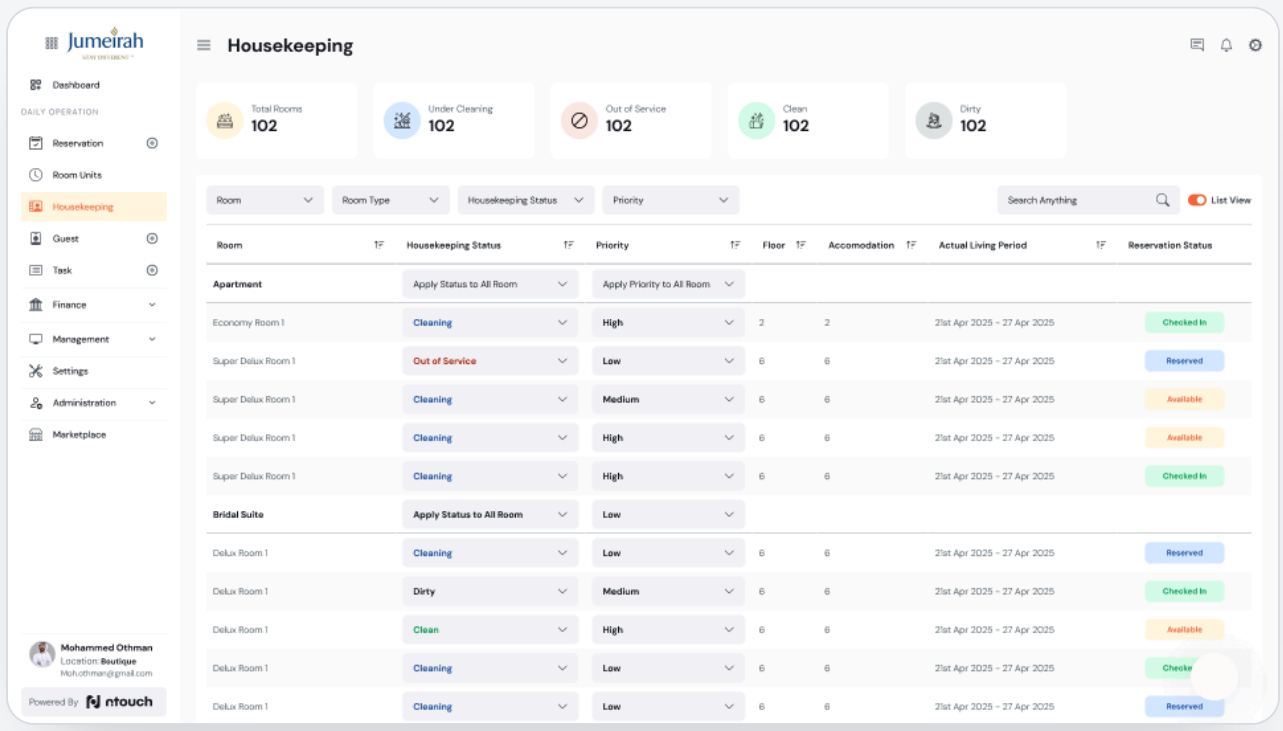
- **Shuttle pattern** moves rooms into a "Selected room" basket with a running total.
- Applied pricing rule (season, high demand, occupancy) shown per room.
- Step two searches existing guests to prefill details and ID.

UNITS

Two statuses on every room card.

- **Occupancy** (available, occupied, checkout, maintenance) and **cleanliness** (clean, not clean) are separate tags.
- Countdown to next check-in or checkout drives priority.
- Inline "Assign" and "Book now" actions on the card.





HOUSEKEEPING

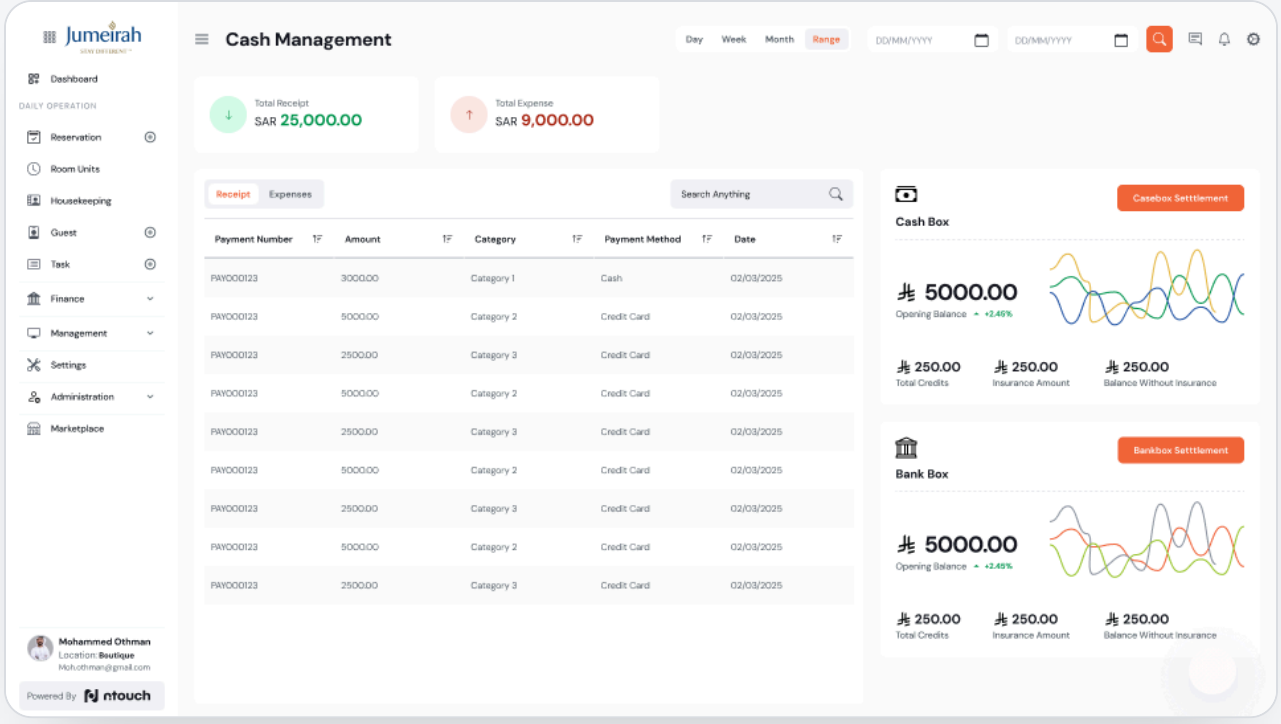
Bulk status changes for a whole floor.

- Status counters across the top: total, under cleaning, out of service, clean, dirty.
- "Apply status to all rooms" per room group for morning sweeps.
- Priority column ties back to reservation status.

FINANCE

Cash box and bank box, side by side.

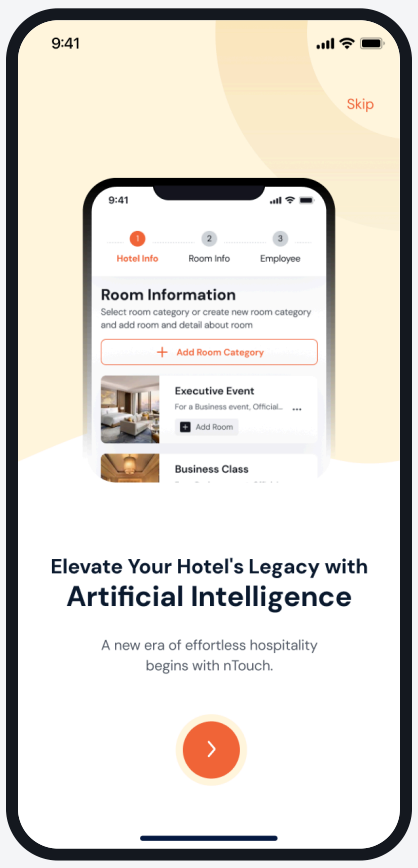
- Receipts and expenses in one table with payment method.
- Separate settlement actions for cash and bank, matching how accountants close the day.
- Amounts in SAR with the new riyal symbol.



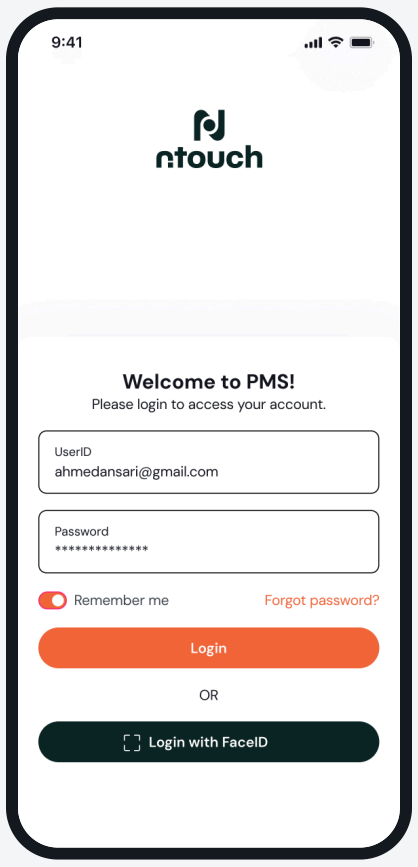
04 · MOBILE

A companion app for people who never sit down.

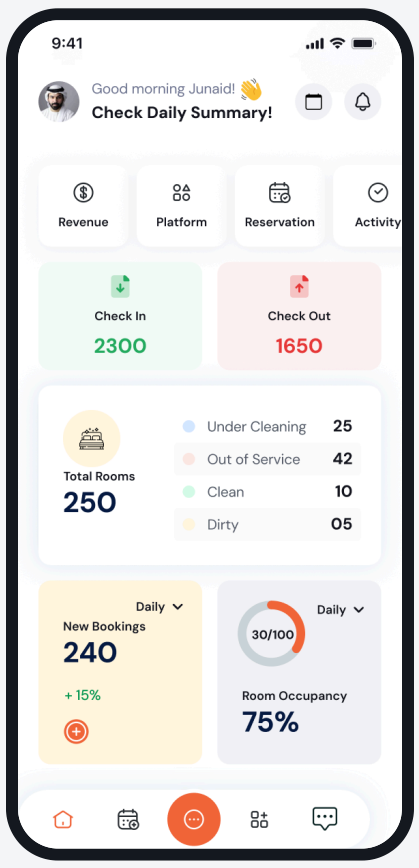
The mobile app is not the web app shrunk. It keeps the jobs people do on the move: checking the day, updating room status, looking up a guest. A central action button in the tab bar opens quick actions.



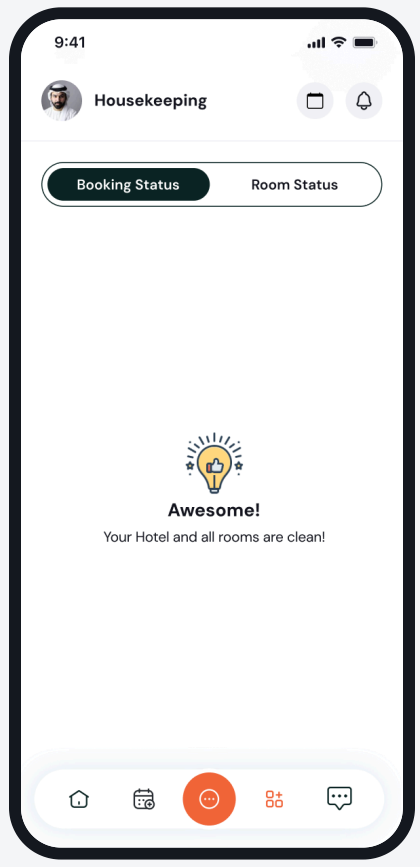
Onboarding
Introduces setup and AI features.



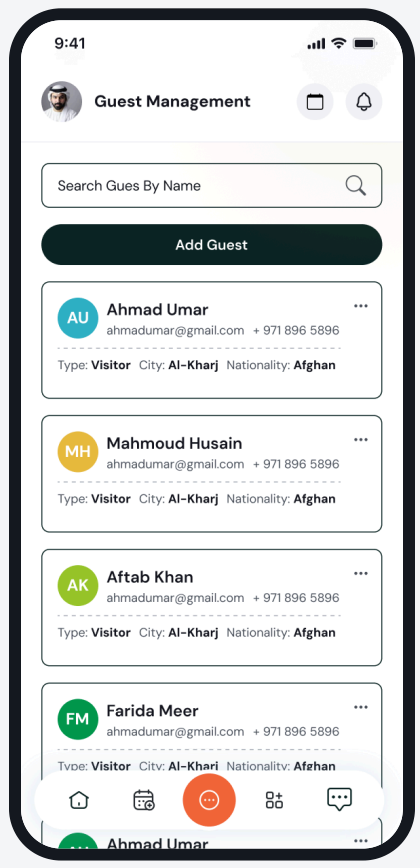
Login
Face ID for shared front-desk phones.



Front desk dashboard
Check-ins, check-outs and room states.



Housekeeping
The zero-pending goal as an empty state.



Guests
Search and add guests on the move.

05 · TEST

Three A/B tests that changed the product.

After moderated usability rounds, we ran A/B tests on the flows with the most traffic in the pilot hotels. Each test had one hypothesis, one primary metric and a fixed run of two weeks.

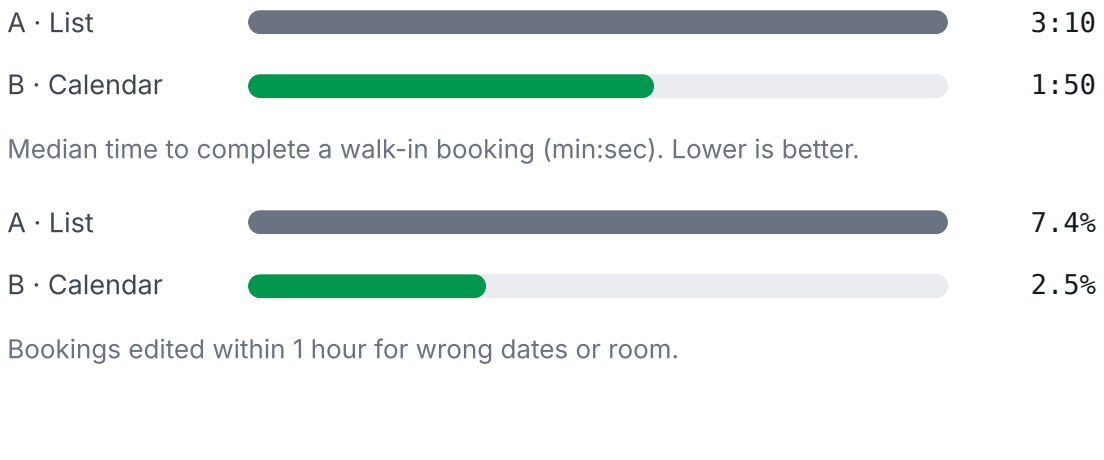
TEST A · RESERVATIONS **B shipped**

Calendar or list as the default view?

If the reservation page opens on the room × night calendar, front-desk staff will create walk-in bookings faster and make fewer date errors, because they can see gaps directly.

A · CONTROL
List view by default, calendar behind a toggle

B · VARIANT
Calendar by default with drag-to-select nights



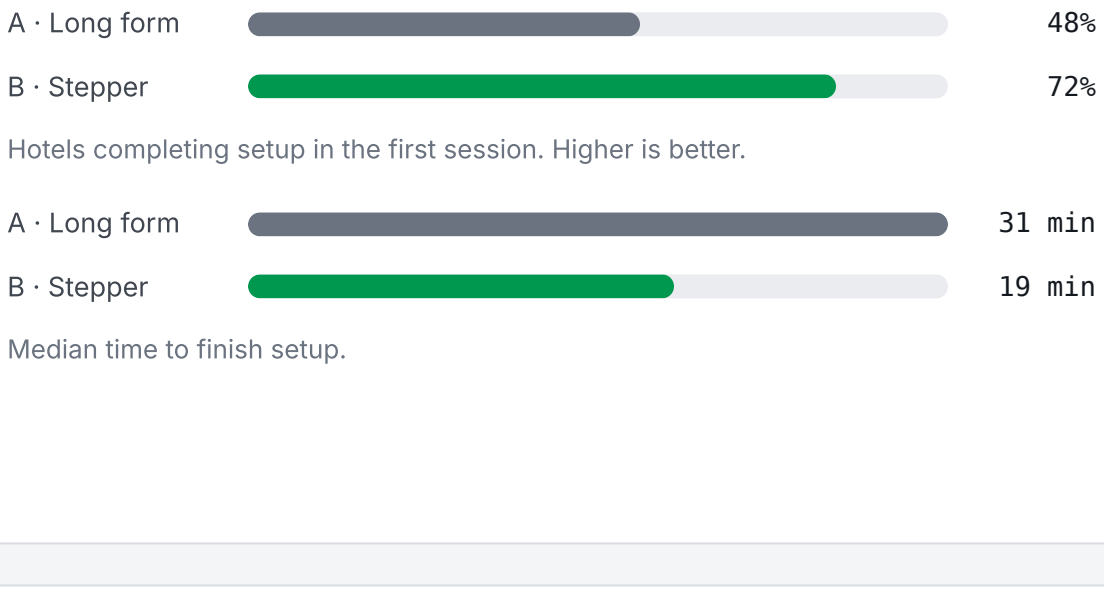
TEST B · ONBOARDING **B shipped**

One long form or a three-step stepper?

If setup is split into Hotel, Room and Employee steps with progress shown, more new hotels will finish setup in their first session.

A · CONTROL
Single scrolling setup form

B · VARIANT
3-step stepper with save per step



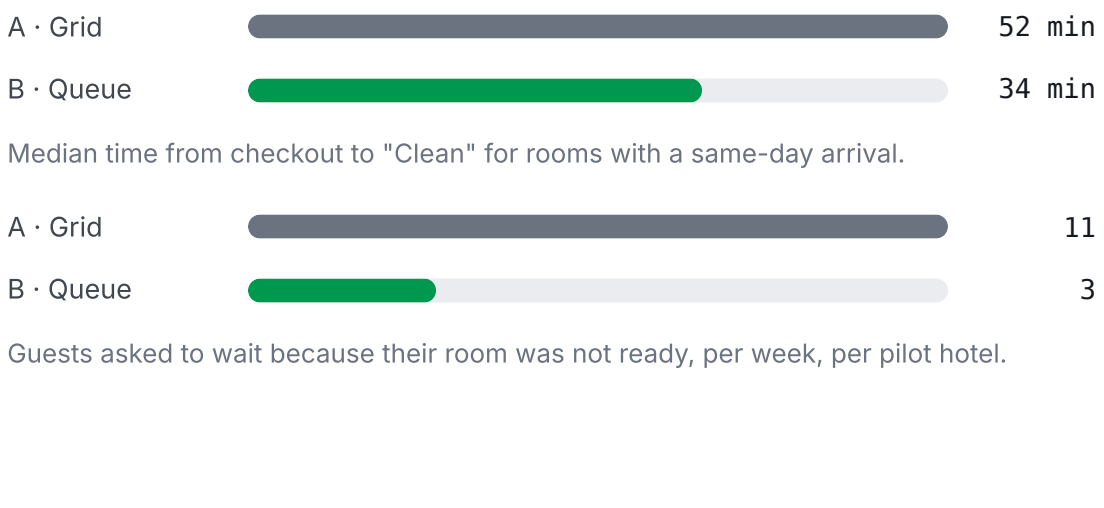
TEST C · HOUSEKEEPING MOBILE **B shipped**

Room status grid or booking-priority queue?

If housekeepers see rooms ordered by the next guest's arrival time, rooms with an arrival today will be ready sooner.

A · CONTROL
Grid of all rooms by floor

B · VARIANT
"Booking status" queue sorted by next arrival



06 · DEVELOPER HANDOFF

Designed to be built, not just presented.

I worked with the engineering team from the design studio onwards, so handoff was a continuation rather than a hand-over. The Figma file was organised by module section, with every flow in order and every state drawn.

Tokens

Color, type and spacing exported as named tokens that mapped 1:1 to CSS variables.

Component specs

Buttons, inputs, dropdowns, chips, price and contact fields as variants with documented props.

States & edge cases

Empty, loading, error, validation, hover and multi-hotel variants drawn for each flow.

Design QA

Weekly review of staging builds against Figma, logged as tickets with screenshots.

```
/* nTouch tokens, as handed to engineering */
--color-primary: #F36437;   --color-primary-hover: #E0542A;   --color-primary-pressed: #C2441C;
--color-success: #98C729;   --color-warning: #E8B93F;         --color-error: #DD331D;   --color-info: #2C68C6;
--text-heading: #14181F;    --text-body: #414750;            --text-placeholder: #8B939F;
--font-family: "DM Sans";   --scale-ratio: 1.2;               --space-unit: 8px;
```

OUTCOME

What changed for the hotels.

Measured across the pilot hotels in the first three months after launch, compared with their previous setup.

−42%
Time to complete a walk-in booking

72%
Of new hotels finish setup in one session

−35%
Checkout-to-clean time for same-day arrivals

1
Place for rooms, guests and money, across every property

Design for the status, not the screen

Once room status had one color and one meaning, every module became easier to design and easier to learn.

Shadowing beats surveys

The most important insight, that everyone was asking "is the room ready?", came from watching a shift, not from asking about one.

Next

Arabic RTL layouts across the web app, AI-assisted pricing suggestions, and guest self check-in.